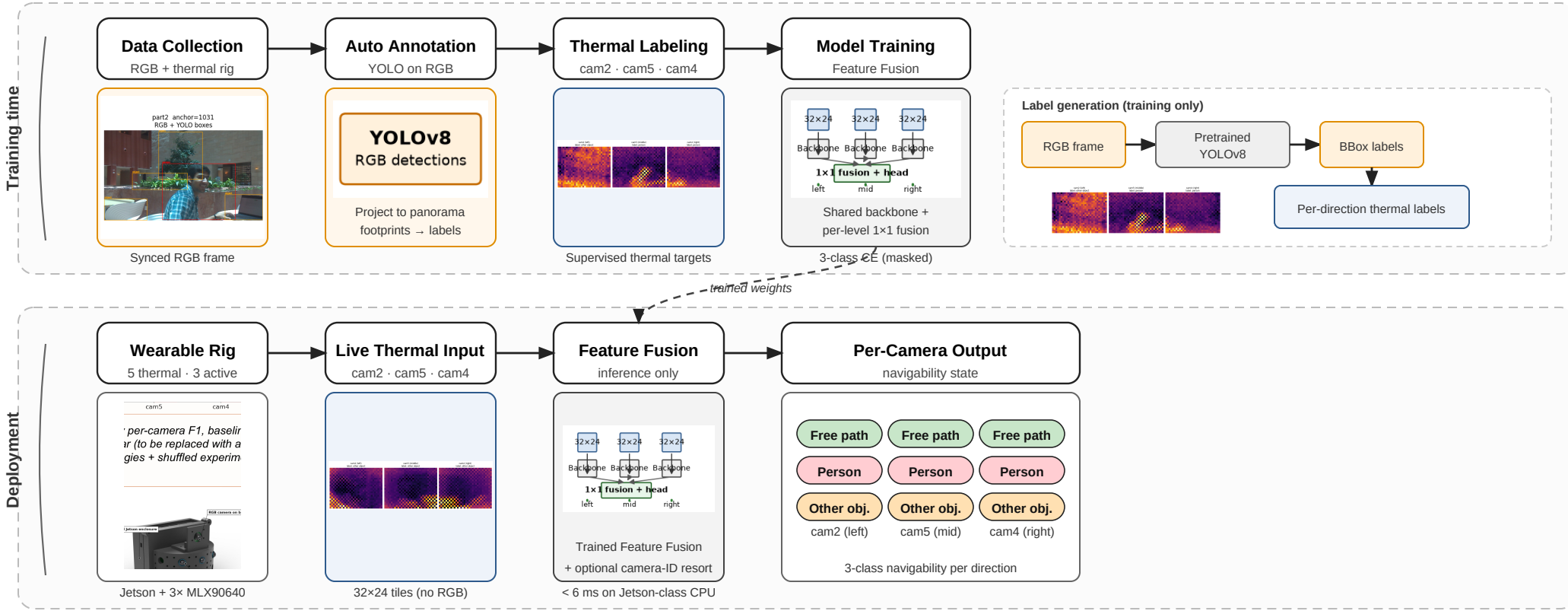

Training time
Data Collection
RGB + thermal rig
Auto Annotation
YOLO on RGB
Thermal Labeling
cam2 · cam5 · cam4
Model Training
Feature Fusion
part2 anchor=1031
RGB + YOLO boxes
YOLOv8
RGB detections
Project to panorama
footprints → labels
32×24
Backbone
1×1 fusion + head
left
mid
right
Shared backbone +
per-level 1×1 fusion
Label generation (training only)
RGB frame
Pretrained
YOLOv8
BBox labels
Per-direction thermal labels
Synced RGB frame
Supervised thermal targets
3-class CE (masked)
trained weights
Deployment
Wearable Rig
5 thermal · 3 active
Live Thermal Input
cam2 · cam5 · cam4
Feature Fusion
inference only
Per-Camera Output
navigability state
cam5
cam4
Free path
Person
Other obj.
cam2 (left)
cam5 (mid)
cam4 (right)
Trained Feature Fusion
+ optional camera-ID resort
3-class navigability per direction
Jetson + 3× MLX90640
32×24 tiles (no RGB)
< 6 ms on Jetson-class CPU